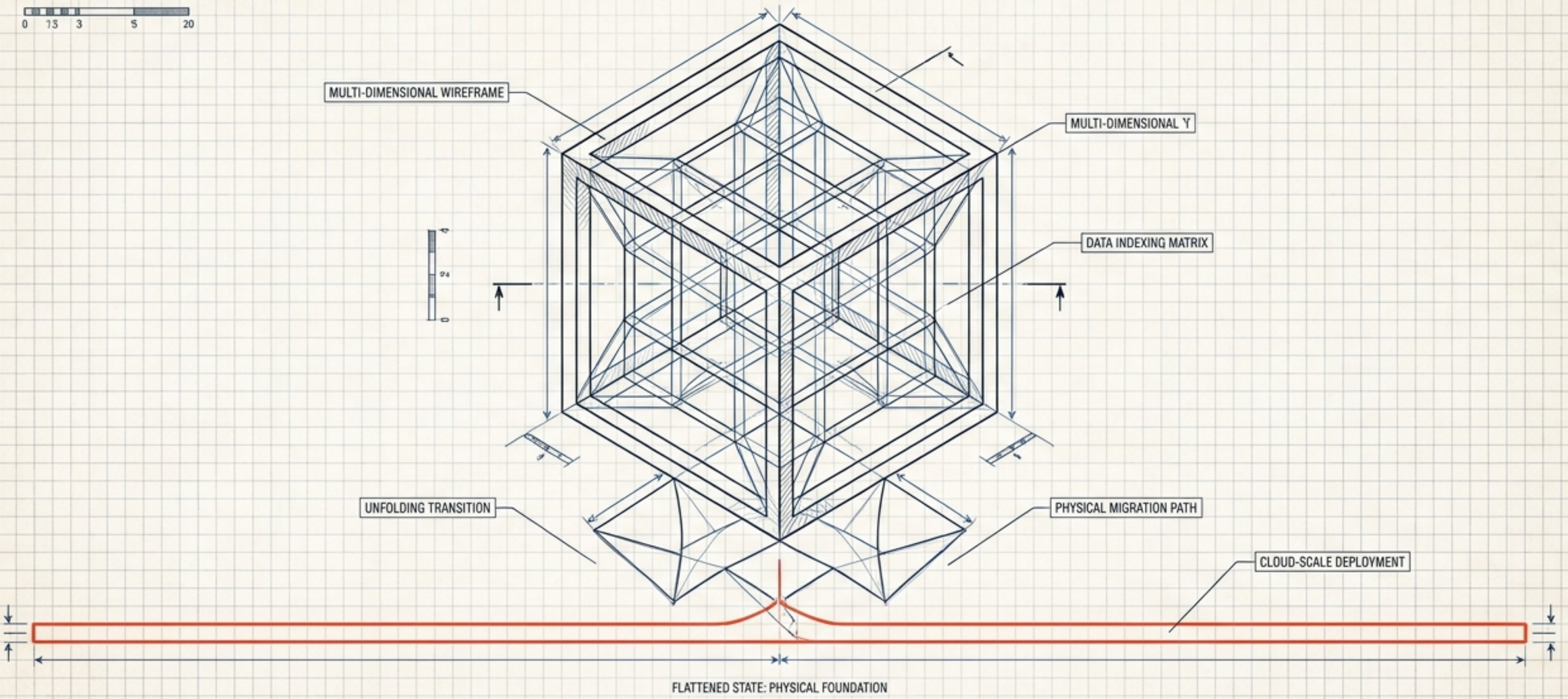
PROJECT: CLOUD ARCHITECTURE REV 2.0

DATE: OCT 26, 2024

SCALE: NTS

THE MASTER PLAN FOR CLOUD-SCALE ARCHITECTURE

FROM PHYSICAL MIGRATIONS TO MULTI-DIMENSIONAL DATA INDEXING



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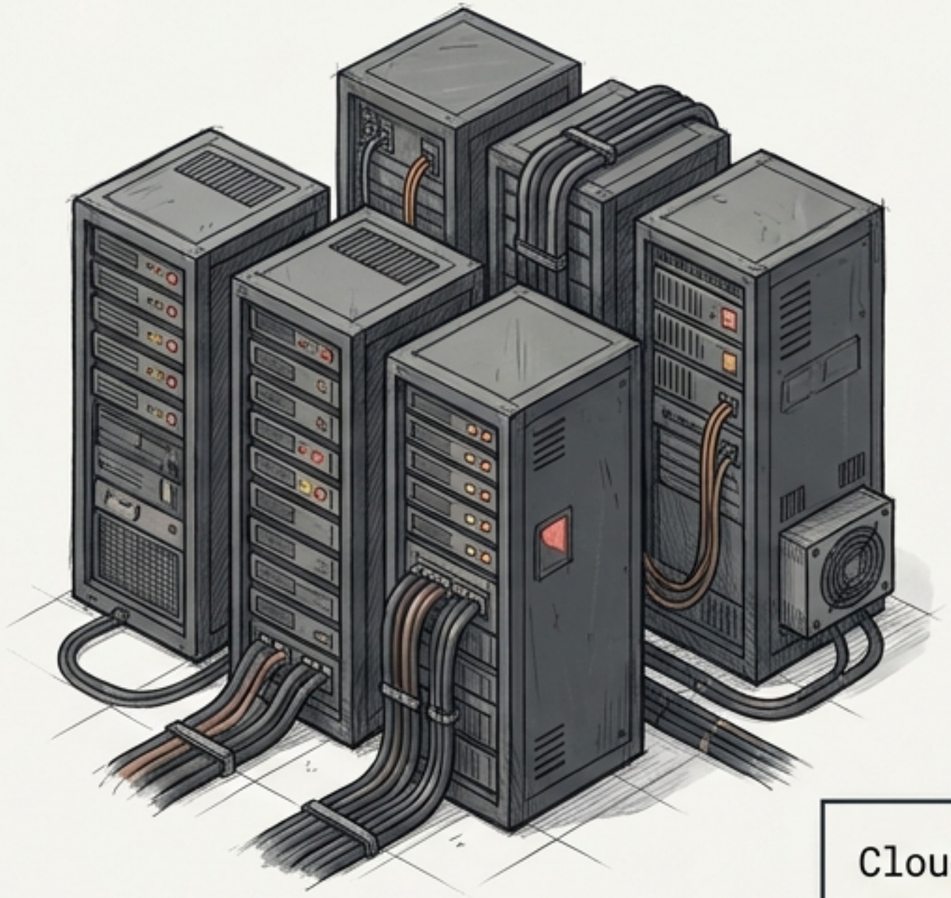
THE OLD PARADIGM



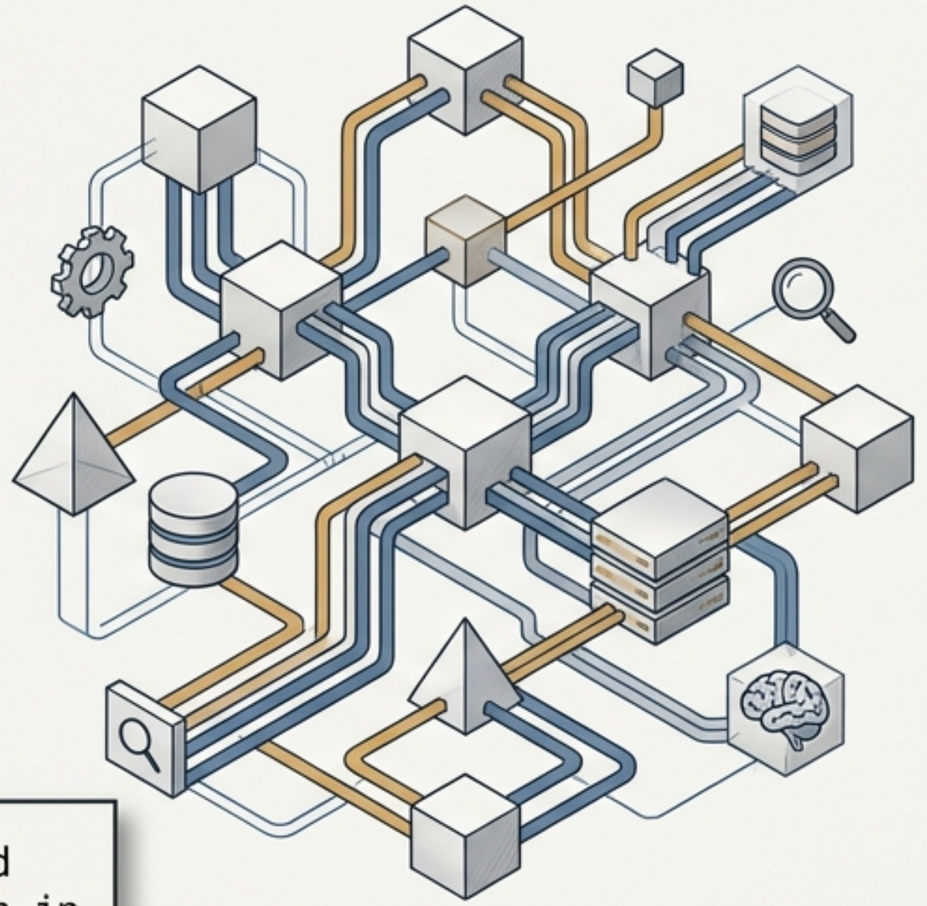
THE OLD PARADIGM

Focus: Where a resource is physically located.

Role: Server Landlords.

> Cloud adoption could generate $3 trillion in EBITDA value by 2030.
>
> — McKinsey & Company

THE CLOUD PARADIGM



THE CLOUD PARADIGM

Focus: What a resource can do.

Role: Architects of Innovation.

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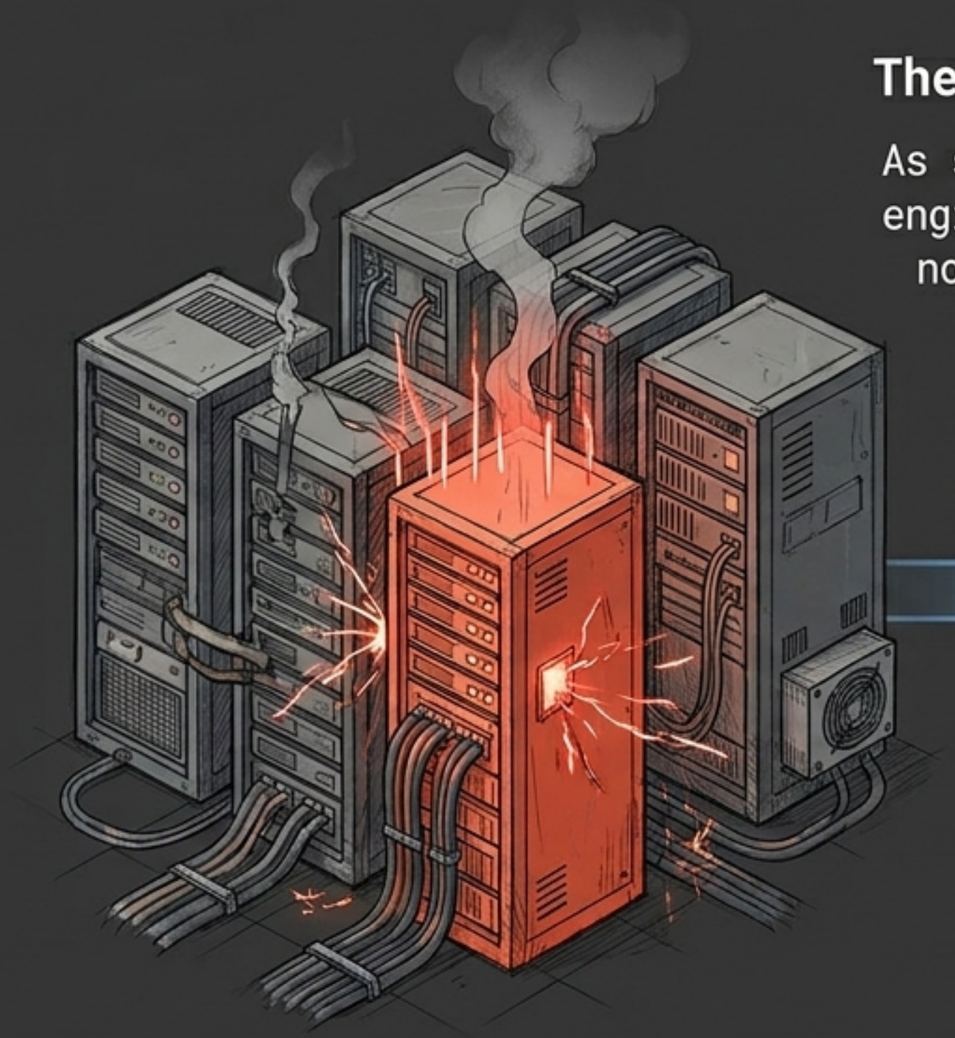
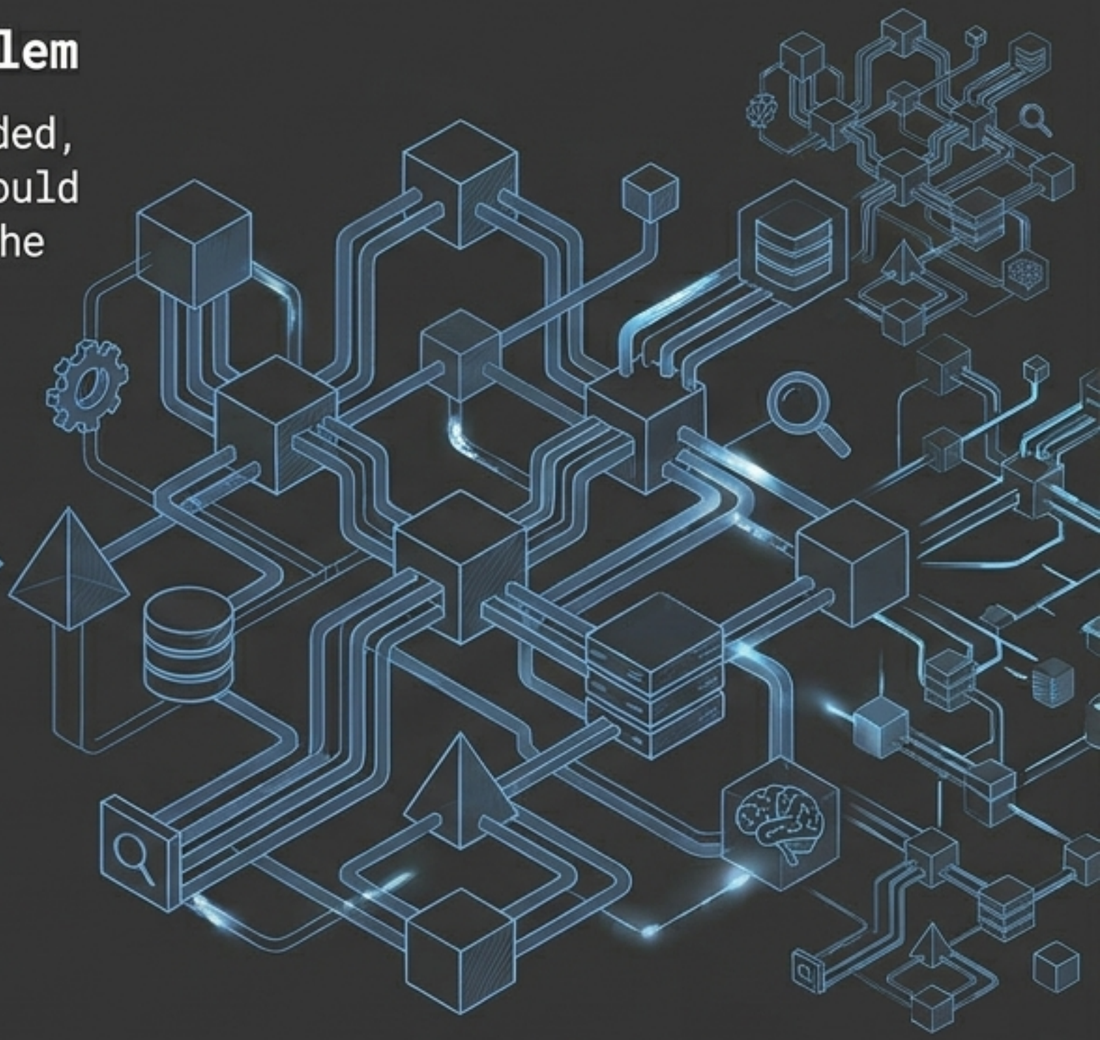


The Growth Problem

As streaming exploded, engineers simply could not have racked the servers fast enough.



The Catalyst (2008)

A single point of failure stops all DVD shipments.

The Result

Transitioning to AWS yielded a 94% reduction in downtime and massive agility, paving the way for billions of streaming hours.

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FINANCIAL IMPACT

Cost Savings

- 15–40% infrastructure savings (BCG)
- 51% reduced operational costs (AWS)

OPERATIONAL ENGINE

- **Scalability:** Instant resource allocation.
- **Security:** Advanced regulatory alignment.
- **Disaster Recovery:** Geo-replication.
- **Accessibility:** Global remote access.

Resource Efficiency

62% increased IT productivity (AWS).

Stop fixing broken servers and start building the next big feature.

ENVIRONMENTAL

Impact Reduction

Energy-efficient shared data centers significantly reduce organizational footprints.

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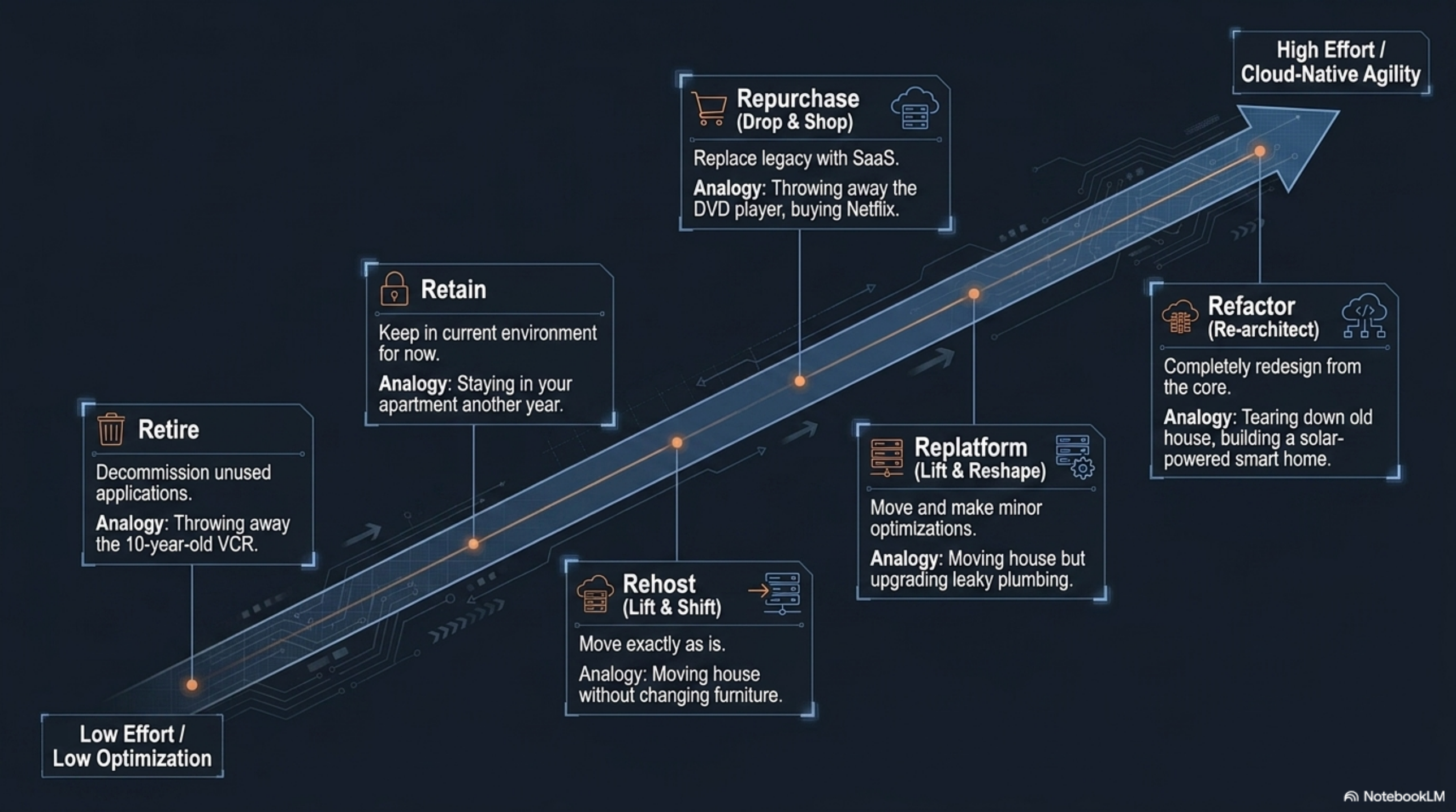
High Effort / Cloud-Native Agility
Repurchase (Drop & Shop)
Replace legacy with SaaS.
Analogy: Throwing away the DVD player, buying Netflix.
Retain
Keep in current environment for now.
Analogy: Staying in your apartment another year.
Refactor (Re-architect)
Completely redesign from the core.
Analogy: Tearing down old house, building a solar-powered smart home.
Retire
Decommission unused applications.
Analogy: Throwing away the 10-year-old VCR.
Replatform (Lift & Reshape)
Move and make minor optimizations.
Analogy: Moving house but upgrading leaky plumbing.
Rehost (Lift & Shift)
Move exactly as is.
Analogy: Moving house without changing furniture.
Low Effort / Low Optimization

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1
1. Assessment & Planning
Analyze infrastructure and select an R strategy.
2
HURDLE: Bandwidth constraints requiring offline petabyte transfers (AWS Snowball).
2. Data Migration
Categorize and move data.
3
HURDLE: Legacy compatibility requiring a Hybrid Cloud approach.
3. Application Migration
Evaluate dependencies and move code.
4
4. Testing & Validation
Functional, security, and User Acceptance Testing (UAT).
5
HURDLE: Culture Shock requiring stakeholder buy-in and training.
5. Optimization & Monitoring
Continuous loop of adjusting resources.

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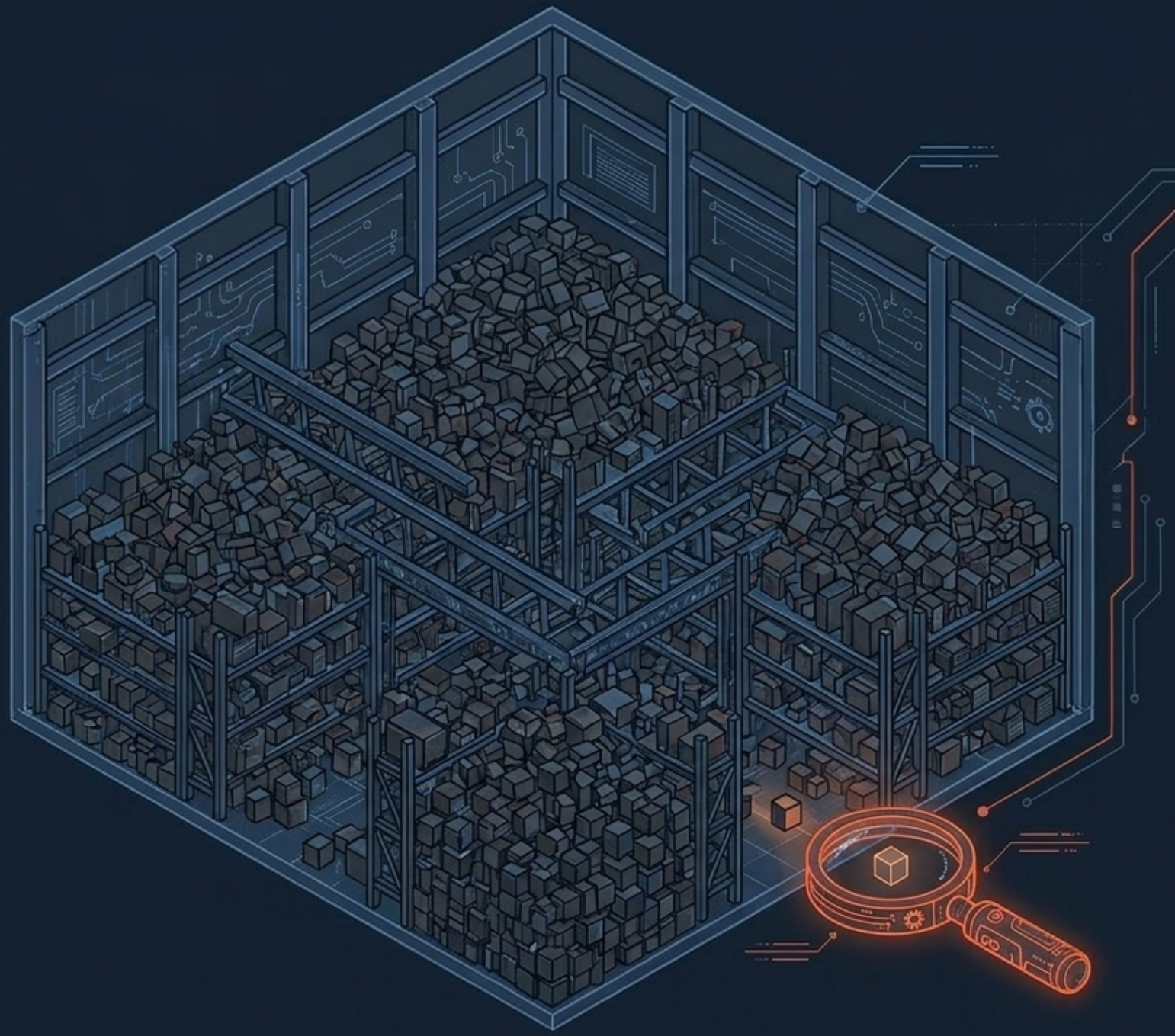

The Secret Sauce of Scalability

Moving to the cloud provides infinite storage. But infinite storage creates an infinite search problem.

Migration is useless if you cannot find what you migrated. The massive agility gained by companies like Netflix is completely dependent on transitioning from **rigid structured hierarchies to decentralized, multi-dimensional indexing.**

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The Evolution of Discovery

Structured Naming	Attribute-Based Naming
Concept: WHERE data is.	**Concept: WHAT** data is.
`Core Identifier`: A unique path/string.	`Core Identifier`: A collection of (attribute, value) pairs.
Uniqueness: Points to a single specific entity.	**Uniqueness:** Can return multiple matching entities.
Use Cases: DNS, file systems.	**Use Cases:** Directory Services, resource discovery.

The 3 Hurdles of Directory Services

1. **Manual Design:** Identifying relevant attributes.
2. **Metadata Consistency:** Ensuring uniform user inputs.
3. **Search Performance:** Avoiding exhaustive lookup bottlenecks.

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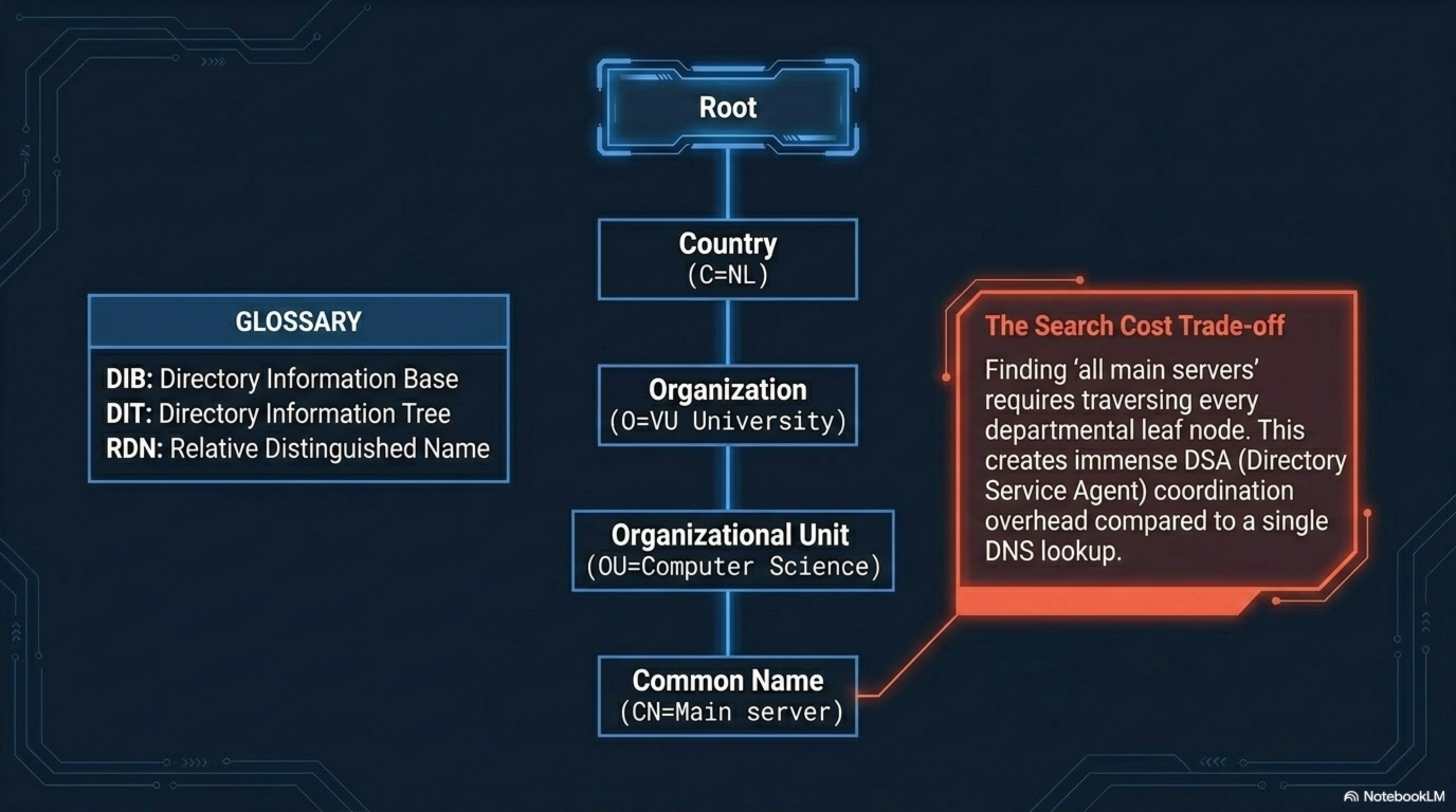
Root
Country
(C=NL)
Organization
(O=VU University)
Organizational Unit
(OU=Computer Science)
Common Name
(CN=Main server)
GLOSSARY
DIB: Directory Information Base
DIT: Directory Information Tree
RDN: Relative Distinguished Name
The Search Cost Trade-off
Finding ‘all main servers’ requires traversing every departmental leaf node. This creates immense DSA (Directory Service Agent) coordination overhead compared to a single DNS lookup.

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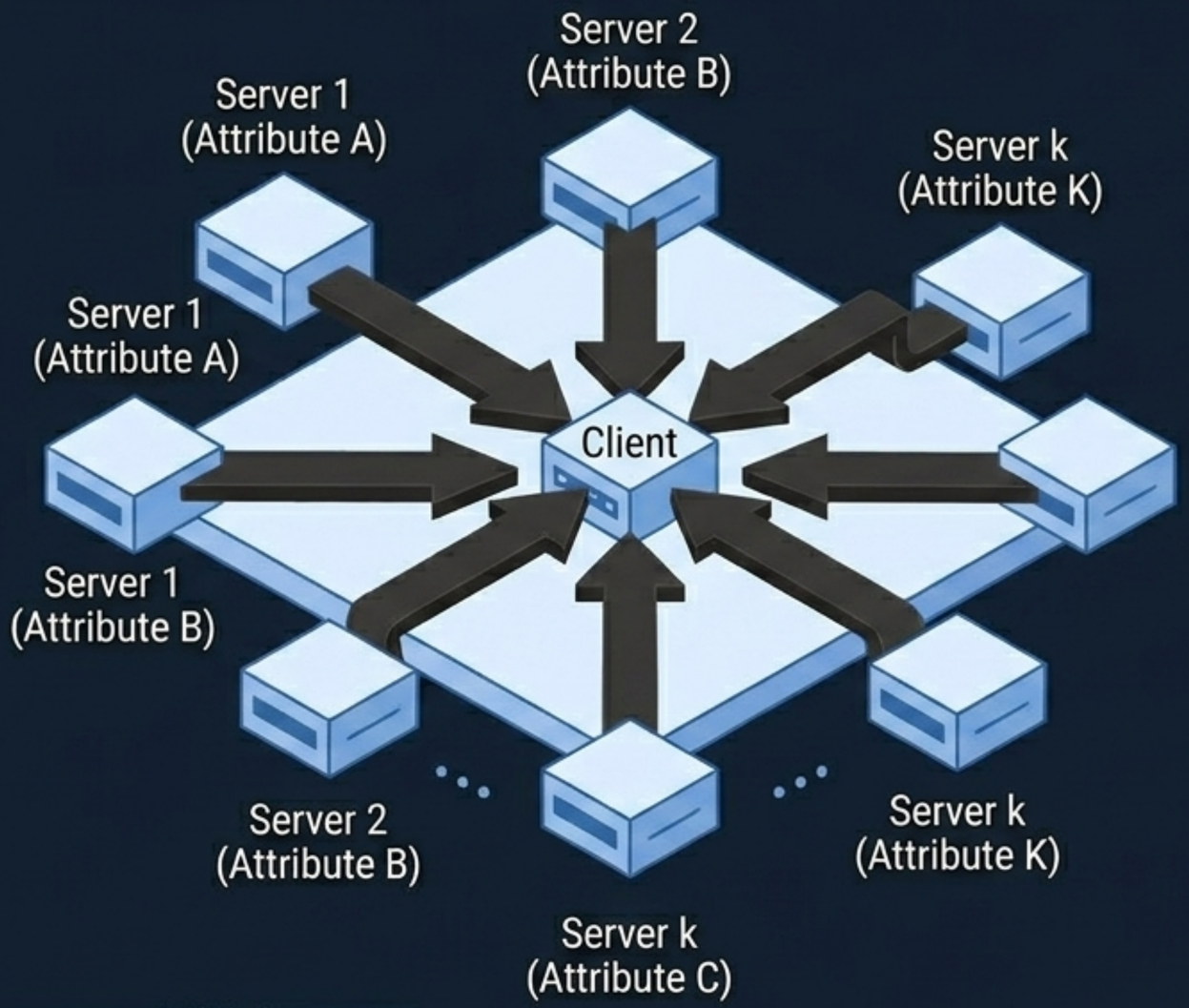
The Pheriby Smith Problem



1. Communication Costs

Querying k attributes requires contacting k distinct servers over the network.

2. Client-Side Processing Load

The ‘Smith’ server returns millions of keys; the client must locally cross-reference them against the ‘Pheriby’ results.

3. Lack of Range Queries

Simple mapping fails at identifying bounds like Price = [$1000–$2500].

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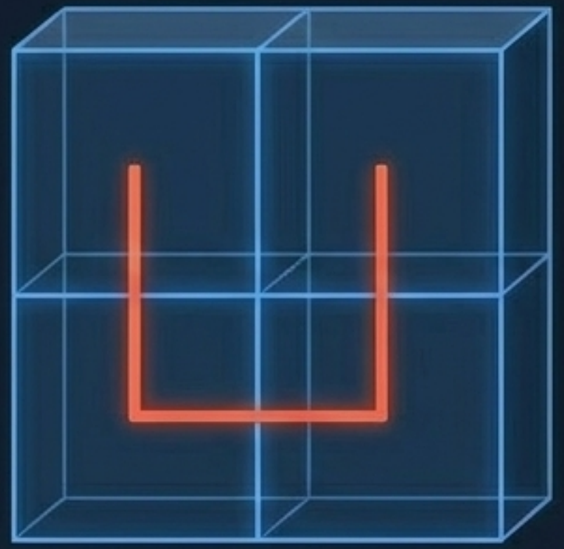
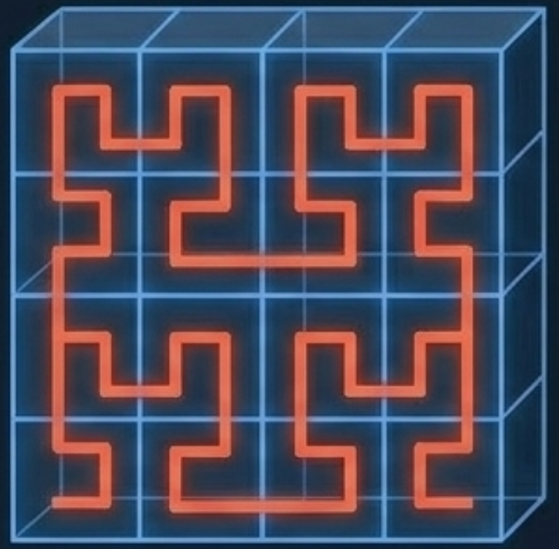
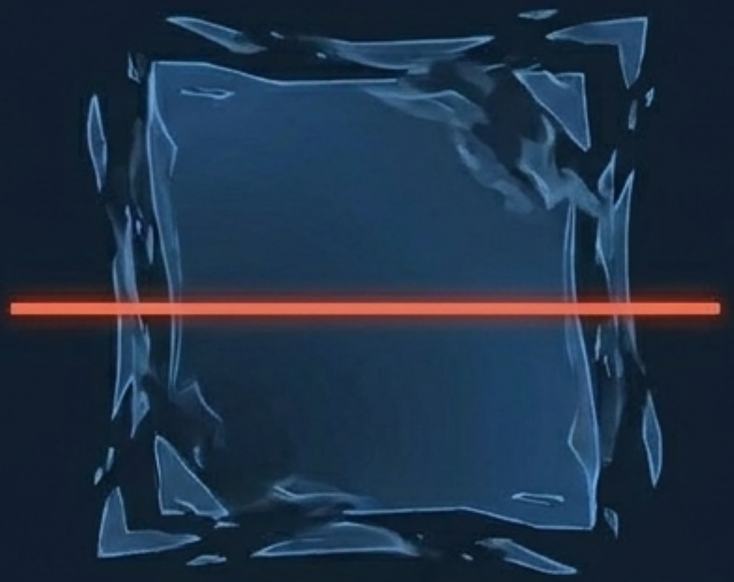
Flattening Space: The Hilbert Curve



Order 1
(4 Quadrants)



Order 2
(16 Squares)



Space Flattening
(1D Line)

Mapping the Scale of Space-Filling Curves
Order 1: 4 Indices
Order 2: 16 Indices
Order 4: 256 Indices
Order 32: 2^64 Indices (The standard for real-world cloud systems).

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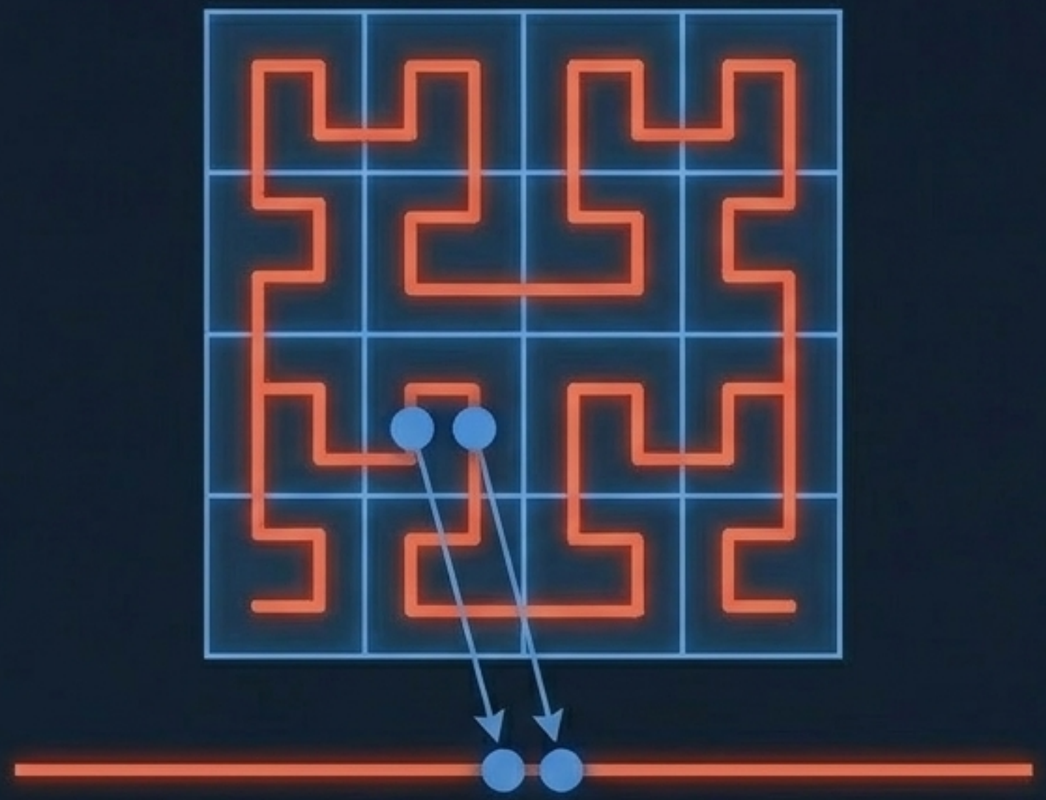
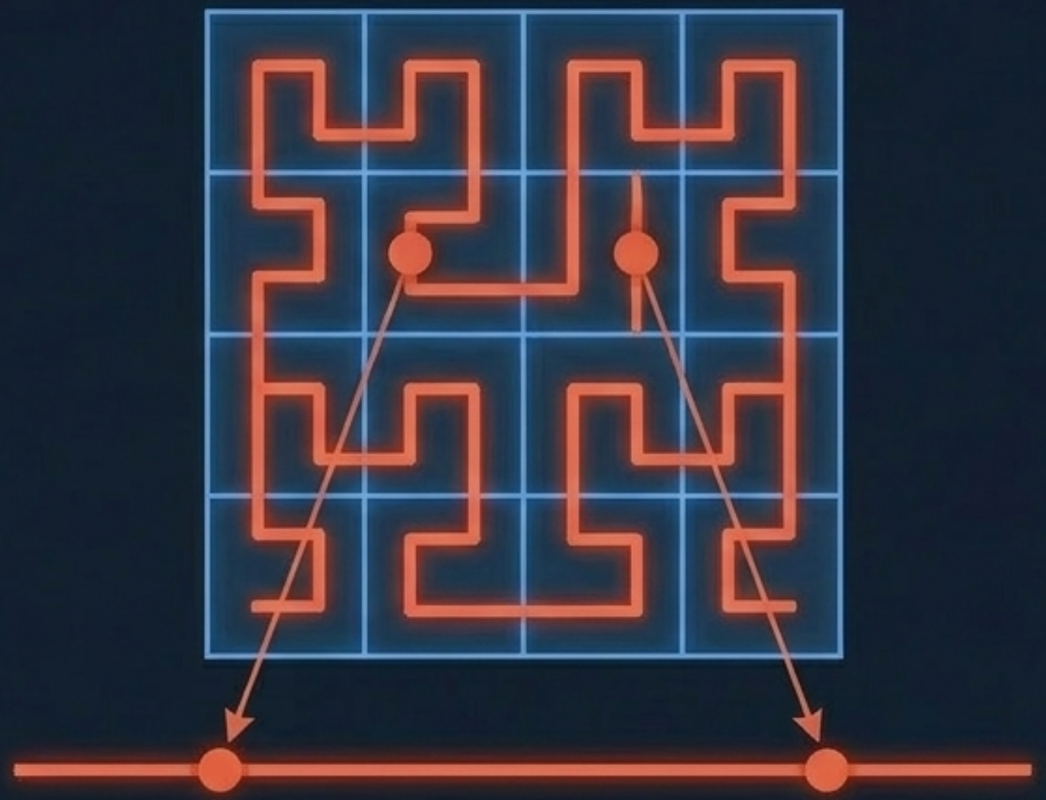
The Magic of Locality



Solves the “Pheriby Smith” problem. A single index server can co-locate related attributes because N-dimensional proximity usually translates to 1D linear proximity.

The Architect’s Warning



Edge Case: Proximity in N-dimensional space does not guarantee proximity on the curve, occasionally necessitating queries to multiple servers.

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Squid Implementation Process

1. Normalization

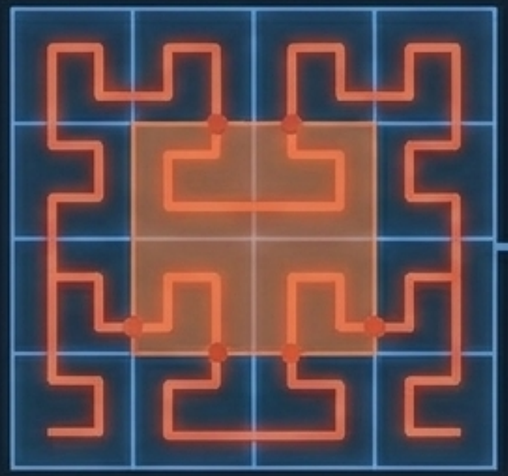
Attribute values are mathematically mapped to a clean interval of `[0, 1)`.

2. Indexing

Coordinates are assigned to a high-order Hilbert curve (`Order 32` or `64`).

3. Storage

Indices are mapped into a Distributed Hash Table (DHT) like a Chord ring. The Hilbert space aligns with Chord's m-bit identifier space.



Range Query Execution

The system routes queries ONLY to the specific curve segments that physically intersect this bounding rectangle.

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Architectural Evaluation: Indexing Alternatives

	Space-Filling Curves (Squid)	Bit Partitioning (SWORD)
Key Construction	Relies entirely on strict mathematical curve geometry.	Uses specific bits for attribute/value and injects random bits to guarantee uniqueness.
Range Queries	Calculates complex curve and rectangle intersections.	Naturally partitions subranges to assigned servers.
Load Balancing Trade-off	Provides highly uniform data distribution across the network.	Risks popular attribute ranges becoming ‘hot’, causing severe server load imbalances.

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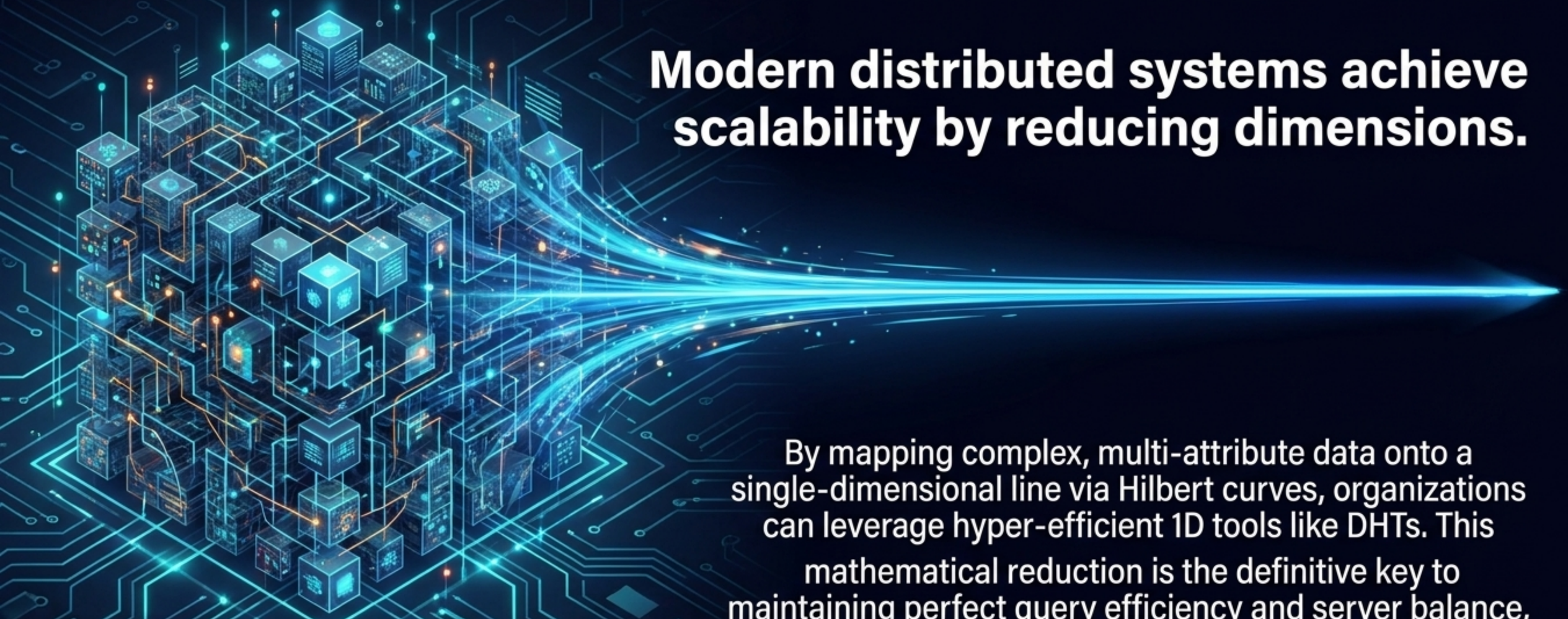
The Synthesis of Dimensionality



Modern distributed systems achieve scalability by reducing dimensions.

By mapping complex, multi-attribute data onto a single-dimensional line via Hilbert curves, organizations can leverage hyper-efficient 1D tools like DHTs. This mathematical reduction is the definitive key to maintaining perfect query efficiency and server balance, unlocking the true, limitless potential of the cloud.

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